

FREDERICTON CDA, NB, Canada

WMO: 716680

Lat: 45.9203N

Lon: 66.6089W

Elev: 35

StdP: 100.90

Time Zone: -4.00 (NAA)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.4	-19.6	-26.8	0.3	-21.1	-24.4	0.4	-18.7	11.1	-4.9	10.1	-5.3	2.4	290	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.3	20.7	27.6	20.0	26.1	19.0	22.2	27.0	21.3	25.8	20.4	24.4	4.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.4	24.6	19.7	14.5	23.4	18.9	13.8	22.5	65.4	27.2	62.0	26.0	58.8	24.4	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.4	-26.5	32.3	3.3	1.3	-28.8	33.2	-30.8	33.9	-32.6	34.7	-35.0	35.6		
(5)			-26.7	23.8	3.2	0.8	-29.0	24.4	-30.9	24.9	-32.7	25.4	-35.0	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-7.8	-7.2	-2.1	5.1	11.6	16.2	20.3	19.2	15.2	8.7	2.3	-4.3		
	DBStd	10.77	6.53	5.90	5.34	3.91	3.57	3.01	2.58	2.55	3.82	4.04	4.80	5.62		
	HDD10.0	2338	552	482	376	153	22	1	0	0	3	72	234	444		
	HDD18.3	4463	810	715	633	397	211	76	11	19	109	297	481	702		
	CDD10.0	1063	0	0	1	6	71	187	318	285	158	33	3	0		
	CDD18.3	147	0	0	0	0	2	13	71	46	14	1	0	0		
	CDH23.3	1176	0	0	2	3	56	128	538	336	108	4	0	0		
	CDH26.7	248	0	0	0	1	12	23	123	63	26	0	0	0		
(14) Wind	WSAvg	3.2	3.6	3.7	3.9	3.6	3.3	2.9	2.7	2.5	2.7	3.1	3.3	3.4		
(15) Precipitation	PrecAvg	1151	105	87	96	86	96	85	88	91	95	97	111	116		
	PrecMax	1520	223	164	194	155	211	199	174	238	271	182	206	206		
	PrecMin	845	20	28	27	20	20	26	29	17	20	21	44	48		
	PrecStd	162	43	35	36	34	47	41	35	51	45	43	39	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	9.1	16.3	21.7	28.1	29.4	31.7	30.5	30.0	23.2	17.2	11.7		
		MCWB	9.9	8.0	10.4	11.8	18.6	19.1	21.8	22.0	21.4	16.2	15.6	10.7		
	2%	DB	6.7	5.2	9.6	17.5	24.5	26.6	29.5	28.4	26.3	19.5	14.4	8.2		
		MCWB	5.8	2.7	4.8	10.5	16.2	18.2	21.0	20.7	19.7	15.4	12.2	7.3		
	5%	DB	3.7	3.1	7.0	14.6	21.5	24.5	27.9	26.7	23.8	17.4	12.2	5.4		
		MCWB	2.5	1.4	4.0	9.2	14.0	17.2	20.3	19.7	18.4	14.6	10.2	4.3		
	10%	DB	1.3	1.3	4.9	12.1	19.0	22.7	26.5	25.2	21.7	15.5	9.8	3.3		
		MCWB	0.3	0.0	2.0	7.8	12.6	16.3	19.6	18.9	17.4	12.7	8.0	2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	7.9	11.2	14.8	19.7	21.4	23.5	23.4	22.2	18.9	16.0	10.8		
		MCDB	10.8	9.2	14.8	18.7	26.4	26.2	28.9	28.3	28.1	20.4	16.8	11.4		
	2%	WB	5.9	3.6	6.3	12.3	17.1	20.1	22.4	22.1	20.8	16.7	13.1	7.4		
		MCDB	6.5	4.6	7.9	15.0	22.3	24.2	27.3	26.5	24.7	18.4	14.0	8.1		
	5%	WB	2.5	1.5	4.2	10.2	15.3	18.7	21.5	21.0	19.5	15.1	10.7	4.5		
		MCDB	3.5	2.7	6.6	13.7	19.5	22.4	26.1	25.1	22.6	17.4	11.9	5.3		
	10%	WB	0.4	0.2	2.4	8.1	13.7	17.5	20.6	20.0	18.2	13.0	8.0	2.2		
		MCDB	1.3	1.2	4.7	11.5	17.6	21.1	24.9	23.4	20.8	14.9	9.6	3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.7	9.4	10.0	11.2	10.3	10.6	10.5	10.7	8.9	7.6	7.4		
		MCDBR	11.6	10.1	11.5	15.4	16.2	14.1	13.0	13.0	12.7	11.8	11.1	9.5		
	5% WB	MCWBR	11.3	8.6	8.4	9.4	8.7	7.3	6.1	6.5	7.2	8.2	9.3	9.3		
		MCWBR	11.6	9.5	10.7	12.8	13.6	11.6	11.1	11.1	11.1	10.2	9.7	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.285	0.316	0.346	0.373	0.405	0.424	0.398	0.369	0.327	0.305	0.286		
		taud	2.456	2.407	2.432	2.425	2.392	2.306	2.274	2.347	2.405	2.527	2.540	2.477		
	Ebn at Noon		844	901	916	914	894	863	840	849	846	842	801	787		
		Edh at Noon	67	86	97	108	115	126	129	116	100	77	63	60		
(44) All-Sky Solar Radiation	RadAvg		1.49	2.36	3.51	4.35	4.99	5.40	5.48	4.85	3.92	2.42	1.42	1.10		
	RadStd		0.16	0.21	0.23	0.41	0.57	0.41	0.44	0.31	0.29	0.23	0.16	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+50	N/A

Nomenclature: See separate page